

Bouncing Seat: An Immersive Virtual Locomotion Interface with LSTM Based Body Gesture Estimation

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ABSTRACT

A pilot study of the bouncing seat system is presented. This system consists of an air cushion and gravicorder. The former enables users to move easier and feel lively motion. The latter and its computing units detect users' body sway and recognize its gestures. Four commands (walk, right turn, left turn, and jump) and associated intuitive body gestures were designed. The estimator based on the multi-timescale LSTM was trained using newly created body gesture dataset and achieved 88% inference accuracy. Results suggest that our system has potential for providing a higher sense of immersion and enjoyment than the joysticks.

Keywords: Virtual reality, immersion, neural network.

Index Terms: Human-centered computing—Human Computer Interaction (HCI) —Interaction paradigms—Virtual reality, Human-centered computing—Human Computer Interaction (HCI) — Empirical studies in HCI

1 INTRODUCTION

Head mount displays have emerged in the past several years and numerous applications dedicated to virtual reality (VR) systems have been launched in the market. Then, not a few applications require users to stay in stationary positions and perambulate in virtual spaces using keyboards or joysticks. In this context, we often hear concerns regarding a lower sense of reality in virtual walking. A reason for sacrificing reality could be the absence of body motion and its sensation. Lack of body-centered interaction causes sensory mismatch, which potentially leads to reducing immersion as well as inducing cybersickness. In this study, the bouncing seat system is proposed, where intuitive body gestures and their passive sensorimotor feedbacks were expected to improve a sense of immersion. Also, the pilot experiments to evaluate user experience quality and cybersickness severity were conducted and their results are reported.

2 RELATED WORK

Many approaches have been researched and devised to emulate legged locomotion in virtual environments. One of the representative methods is redirected walking. It allows users to perform virtual walking in larger space than physical space by unnoticeably manipulating users' walking trajectories in virtual space. Another known approach is the repositioning system like omnidirectional treadmills. It only uses limited small workspace but enables users to physically behave in the same manner as real walking. The walk-in-place (WIP) method is also known, which utilizes the



Figure 1: Bouncing seat system and VR application.

walking-like gestures such as marching in place or arms swinging. Especially for the seated WIP methods [1], we can find interesting interfaces like sliding available shoes or ankle-actuated seat [2]. We also propose the seated WIP interface using the air cushion and gravicorder as shown in Figure 1.

3 METHODS

Figure 1 shows our system. To improve reality of virtual locomotion, the commercially available air cushion was used. Moderately enhanced body motion was expected to improve a sense of reality. In addition, the commercially available gravicorder was placed underneath. It had weight sensors in four corners and detected the center of gravity sway of sitting users. According to the time-series body sway information, the body gestures were estimated, and associated control commands were invoked to the VR application. Figure 2 illustrates the block diagram of our system.

3.1 Intuitive body gestures

As a pilot study, following four commands were implemented.

1. Walk: Continuously swinging your waist to the left and right enables moving forward at a constant velocity.
2. Right turn: Shifting your weight to the right side (as putting your weight on your right buttock) for a while enables continuous right turn at a constant angular velocity.
3. Left turn: The same as right turn but applying for the left.
4. Jump: Impulsively pushing the cushion downward (like trying to jump with your hips) enables jumping action.

Figure 3 indicates the example waveforms of each body gesture.

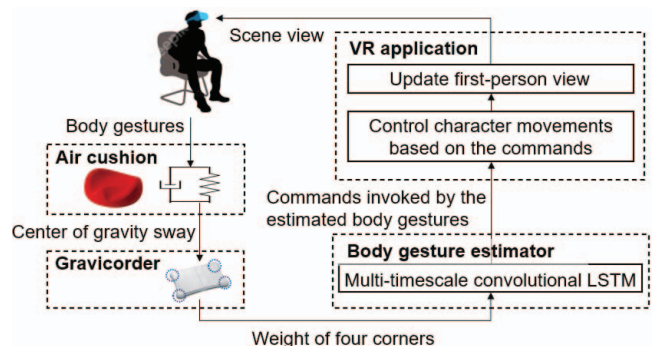


Figure 2: The total system structure.

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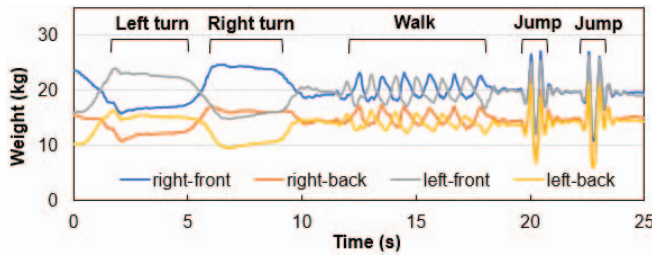


Figure 3: The example waveform of each body gesture.

3.2 Body gesture estimator

For recognizing these four body gestures, the multi-timescale convolutional LSTM (MTSC-LSTM) was applied. It is known that the MTSC-LSTM is superior to distinguishing the similar periodic gestures like nod, jerk, tick, up and down [3]. Figure 4 illustrates the structure of our MTSC-LSTM. We utilized 4 s and 1 s timescales. Dataset was newly created based on the time-series weight variation data during the body gestures of the five subjects. PyTorch based implementation achieved 88 % inference accuracy.

4 EVALUATION

To investigate functionalities of the proposed system and evaluate improvements of reality, we created a VR application where subjects can virtually walk about in a medieval village (Figure 1).

Apparatus: The commercially available VR headset (PSVR), air cushion (Cuvilady) and gravicorder (WiiFit) were used.

Participants: Three male and two female subjects with normal or corrected-to-normal vision, aged from 20 to 55, were participated.

Tasks: Participants were instructed to follow the same route of approximately ten min virtual walk using the conventional joysticks and the proposed bouncing seat. The traveling route contained 20 right turns, left turns and jump events, respectively.

Procedure: The experiments were proceeded in the order of followings: (1) ten min guidance of system apparatus, control methods and detailed tasks, (2) one min practice of traveling using joysticks, (3) five min break, (4) about ten min task performance using joysticks, (5) one hour break, (6) one min practice traveling using bouncing seat, (7) five min break, (8) about ten min task performance using bouncing seat, and (9) answer 20 questions. Questions were selected from the immersive experience questionnaire (IEQ) [4], the short version of user experience questionnaire (UEQS) [5] and the simulator sickness questionnaire (SSQ) [6].

5 RESULTS

We grouped 20 questions into following five categories:

- Easiness: IEQ-3, 16, UEQS-1, 2, 3, 4
- Enjoyment: IEQ-13, 29, UEQS-5, 6, 7
- Immersion: IEQ-6, 14, 15, 19
- Fatigue: SSQ-2
- Nausea: SSQ-7, 8, 15, 16.

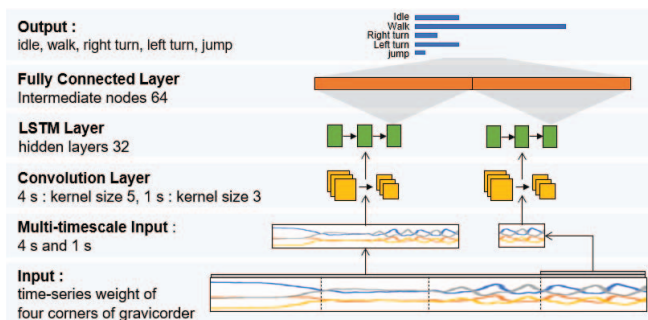


Figure 4: Multi-timescale convolutional LSTM.

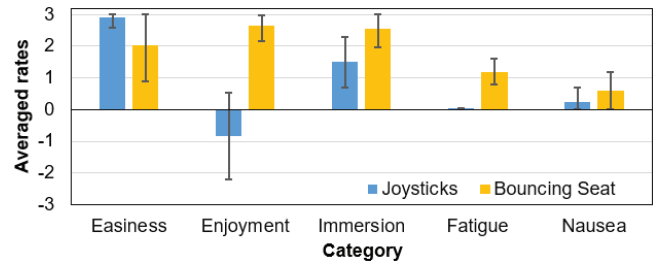


Figure 5: The averaged rates of each category.

Here, the indicated symbols are associated item ID in the original questionnaires. Participants rated each item on a 7-point Likert scale (from -3 to +3) for A, B and C, and on a 4-point Likert scale (from 0 to +3) for D and E. Figure 4 shows the averaged rates of each category with using the joysticks and the bouncing seat.

6 DISCUSSION

Results suggest first trends as follows:

- although the joysticks seem to be easier to operate, functionalities of the bouncing seat are also supportive and efficient,
- ours seems to be more enjoyable and immersive, but also
- risks of triggering cybersickness seem to be higher.

According to the separated interviews, the most common reason for difficulties of operating our system was the timing control in finishing turns. Under- or over-rotation will require additional corrections so that more accurate angle control in finishing turns would be important for better usability. Variable angular velocity control according to the amount of weight shift may contribute to the more accurate adjustment of turning angles. On the other hand, the worse fatigue and nausea scores of our system imply the higher potential risks for cybersickness than the joysticks. It could be improved by supporting on the back and/or elbows and forearms like we applied in [7]. An improved sense of stability could be the solution for reducing potential triggers of cybersickness.

7 CONCLUSION

Results of the pilot experiments suggest that the proposed bouncing seat system has potential for providing a higher sense of immersion and enjoyment than the conventional joysticks in virtual locomotion. Although the body gesture estimator based on the MTSC-LSTM accomplished stable command operation, more angular-accurate control will contribute for better useability. Mitigation of potential risks for cybersickness is our future work.

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